Reproduction Guide

Package layout

The manuscript and reproducibility materials are organized as follows:

- 01_Main_Manuscript.pdf: manuscript PDF.
- 02_Cover_Letter.pdf: cover letter for the editorial system.
- 03_LaTeX_Source.zip: LaTeX source used to compile the manuscript, if requested by the journal system.
- 04_Supplementary_Information.pdf: overview of all supplementary files.
- 05_Reproduction_Guide.pdf: this document.
- 06_Data_Input/: source manifests and derived input data.
- 07_Data_Results/: validation, result, output-summary, compact model-evidence, figure-source, and runtime-provenance data.
- 08_Code/: scripts for reconstruction, reproduction, validation, runtime checks, and figures.
- 10_Figures_optional/: separate figure files for upload only if the journal system requests them.

External source data

Original public image pixels are not redistributed in the supplementary files or in the repository. The manuscript uses MIPDB as a public source dataset. Obtain the source images from the public MIPDB portal cited in the manuscript, then use the supplied source manifests and derived label tables to reconstruct the working label files.

The source-data audit files are in:

- 06_Data_Input/04_S12a_MIPDB_source_manifests.zip
- 06_Data_Input/05_S12b_label_only_derived_input_tables.zip

Reconstruct derived label files

1. Unzip 06_Data_Input/05_S12b_label_only_derived_input_tables.zip.
2. Unzip 08_Code/01_S12d_reproducibility_code.zip.
3. Run the label reconstruction script from the directory that contains the unzipped derived label tables:

```
python label_reconstruction/reconstruct_yolo_label_files.py \
--tables derived_yolo_label_tables \
--out-root derived_yolo_label_datasets_reconstructed
```

4. Place the public source images into the reconstructed dataset folders according to the supplied image lists and manifests.

Reproduce the main analyses

After reconstructing labels and linking the public images, use the scripts in 08_Code/01_S12d_reproducibility_code.zip. The command records, environment summary, seeds, model summaries, and manuscript table sources are provided in 07_Data_Results/08_S12c_result_summaries_and_runtime_tables.zip.

Recommended audit sequence:

1. Inspect 06_Data_Input/ to confirm source manifests, camera mapping, scale sources, and derived input tables.
2. Reconstruct label files with the script above.
3. Use the command records and configuration summaries in S12c_result_summaries_and_runtime_tables.zip to rerun the staged detection, phenological routing, height-related measurement, leaf-view calibration, and system-output generation steps.
4. Compare regenerated summary tables with the files in 07_Data_Results/.

The expected software environment is a standard Python scientific stack with pandas, numpy, opencv-python, scikit-learn, and ultralytics, plus R for figure reproduction where applicable. The exact environment and command records are included in the result-summary archive.

Reproduce the small same-view validation

The small ruler-anchored fixed-view validation is separated into data/results and scripts:

- Data and results: 07_Data_Results/07_S3c_validation_data_and_results.zip
- Scripts: 08_Code/02_S3c_validation_scripts.zip

Unzip both archives into the same working directory and run the scripts from the validation root. The scripts materialize stalk-height annotations, draw annotation references, and compute same-view height-error summaries.

Reproduce figures

Figure source tables are supplied in 07_Data_Results/15_S7_figure_source_tables.zip. Figure-generation scripts are supplied under figure_scripts/ inside 08_Code/01_S12d_reproducibility_code.zip. If the journal requests separate figures during submission, upload files from 10_Figures_optional/. Otherwise the manuscript PDF already contains the figures.

What to upload

For the journal submission system, upload the manuscript PDF, cover letter PDF, required declaration files, supplementary information PDF, and LaTeX source files if requested. The reproduction guide, derived input data, validation/result data, and code files are provided through the public project repository listed in the manuscript Data and Code Availability section. Optional separate figure files should be uploaded only if the journal system requests them. Keep online-form notes and local backups outside the submission.